22542695 PREDICTION OF FOOTBALL MATCH RESULTS USING MACHINE LEARNING MODELS(DATA EXPLORATION)-PART1

1. UPLOAD ESSENTIAL LIBRARIES AND THE EPL DATA

In [1]:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import LabelEncoder, OneHotEncoder, Standar
from scipy.stats import poisson
from sklearn.decomposition import PCA
from sklearn.metrics import mean_squared_error, mean_absolute_error, r
from sklearn.metrics import balanced_accuracy_score
from sklearn.linear_model import LinearRegression
from sklearn.svm import SVR
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.neighbors import KNeighborsRegressor
from sklearn.neural_network import MLPRegressor
import os
import matplotlib.pyplot as plt
%matplotlib inline
%config InlineBackend.figure_format = 'retina'
import seaborn as sns
sns.set(
    style='ticks',
    context='talk',
    font_scale=0.8,
    rc={'figure.figsize': (8,6)}
)
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import PolynomialFeatures
from sklearn.feature_selection import SelectFromModel
from sklearn.feature_selection import SelectKBest, f_regression, f_cla
from sklearn.feature_selection import SequentialFeatureSelector
from sklearn.feature_selection import RFECV
from sklearn.model_selection import StratifiedKFold
from sklearn.model_selection import KFold
from sklearn.linear_model import LinearRegression, Ridge, Lasso, Elast

from functools import partial


rmse = partial(mean_squared_error, squared=False)
from sklearn.model_selection import cross_val_score
from sklearn.decomposition import PCA
from sklearn.decomposition import TruncatedSVD
from sklearn.pipeline import make_pipeline
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
from sklearn.model_selection import cross_val_score

!pip install shap
!pip install -q shap
import shap
shap.initjs()

import random
seed = 42
random.seed(seed)
np.random.seed(seed)

!pip install pandas seaborn matplotlib

```

```
60
```

```
C:\Users\chint\anaconda3\lib\site-packages\numpy\_distributor_init.py:30:
UserWarning: loaded more than 1 DLL from .libs:
C:\Users\chint\anaconda3\lib\site-packages\numpy\.libs\libopenblas.el2c6p
le4zyw3eceviv3oxxgrn2nrfm2.gfortran-win_amd64.dll
C:\Users\chint\anaconda3\lib\site-packages\numpy\.libs\libopenblas.GK7GX5
KEQ4F6UYO3P26ULGBQYHGQO7J4.gfortran-win_amd64.dll
  warnings.warn("loaded more than 1 DLL from .libs:"

Requirement already satisfied: shap in c:\users\chint\anaconda3\lib\site-
packages (0.42.1)
Requirement already satisfied: numba in c:\users\chint\anaconda3\lib\site
-packages (from shap) (0.55.1)
Requirement already satisfied: pandas in c:\users\chint\anaconda3\lib\sit
e-packages (from shap) (1.4.2)
Requirement already satisfied: tqdm>=4.27.0 in c:\users\chint\anaconda3\l
ib\site-packages (from shap) (4.64.0)
Requirement already satisfied: slicer==0.0.7 in c:\users\chint\anaconda3
\lib\site-packages (from shap) (0.0.7)
Requirement already satisfied: packaging>20.9 in c:\users\chint\anaconda3
\lib\site-packages (from shap) (21.3)
Requirement already satisfied: cloudpickle in c:\users\chint\anaconda3\li
b\site-packages (from shap) (2.0.0)
Requirement already satisfied: numpy in c:\users\chint\anaconda3\lib\site
-packages (from shap) (1.22.4)
Requirement already satisfied: scikit-learn in c:\users\chint\appdata\roa
ming\python\python39\site-packages (from shap) (1.3.0)
Requirement already satisfied: scipy in c:\users\chint\anaconda3\lib\site
-packages (from shap) (1.7.3)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in c:\users\chint
\anaconda3\lib\site-packages (from packaging>20.9->shap) (3.0.4)
Requirement already satisfied: colorama in c:\users\chint\anaconda3\lib\s
ite-packages (from tqdm>=4.27.0->shap) (0.4.4)
Collecting numpy
  Using cached numpy-1.21.6-cp39-cp39-win_amd64.whl (14.0 MB)
Requirement already satisfied: llvmlite<0.39,>=0.38.0rc1 in c:\users\chin
t\anaconda3\lib\site-packages (from numba->shap) (0.38.0)
Requirement already satisfied: setuptools in c:\users\chint\anaconda3\lib
\site-packages (from numba->shap) (61.2.0)
Requirement already satisfied: pytz>=2020.1 in c:\users\chint\anaconda3\l
ib\site-packages (from pandas->shap) (2021.3)
Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\chint\a
naconda3\lib\site-packages (from pandas->shap) (2.8.2)
Requirement already satisfied: six>=1.5 in c:\users\chint\anaconda3\lib\s
ite-packages (from python-dateutil>=2.8.1->pandas->shap) (1.16.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\chint\ana
conda3\lib\site-packages (from scikit-learn->shap) (3.2.0)
Requirement already satisfied: joblib>=1.1.1 in c:\users\chint\appdata\ro
aming\python\python39\site-packages (from scikit-learn->shap) (1.2.0)
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 1.22.4
    Uninstalling numpy-1.22.4:
      Successfully uninstalled numpy-1.22.4

ERROR: Could not install packages due to an OSError: [WinError 5] Access
is denied: 'C:\\Users\\chint\\AppData\\Local\\Temp\\pip-uninstall-hr9ry__
4\\libopenblas.el2c6ple4zyw3eceviv3oxxgrn2nrfm2.gfortran-win_amd64.dll'
Consider using the `--user` option or check the permissions.
```

```
Requirement already satisfied: pandas in c:\users\chint\anaconda3\lib\sit
e-packages (1.4.2)
Requirement already satisfied: seaborn in c:\users\chint\anaconda3\lib\si
te-packages (0.11.2)
Requirement already satisfied: matplotlib in c:\users\chint\anaconda3\lib
\site-packages (3.5.1)
Requirement already satisfied: pytz>=2020.1 in c:\users\chint\anaconda3\l
ib\site-packages (from pandas) (2021.3)
Requirement already satisfied: python-dateutil>=2.8.1 in c:\users\chint\a
naconda3\lib\site-packages (from pandas) (2.8.2)
Requirement already satisfied: numpy>=1.18.5 in c:\users\chint\anaconda3
\lib\site-packages (from pandas) (1.21.6)
Requirement already satisfied: scipy>=1.0 in c:\users\chint\anaconda3\lib
\site-packages (from seaborn) (1.7.3)
Requirement already satisfied: pillow>=6.2.0 in c:\users\chint\anaconda3
\lib\site-packages (from matplotlib) (9.0.1)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\chint\anacon
da3\lib\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in c:\users\chint\anaconda3\l
ib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\chint\anacon
da3\lib\site-packages (from matplotlib) (4.25.0)
Requirement already satisfied: pyparsing>=2.2.1 in c:\users\chint\anacond
a3\lib\site-packages (from matplotlib) (3.0.4)
Requirement already satisfied: packaging>=20.0 in c:\users\chint\anaconda
3\lib\site-packages (from matplotlib) (21.3)
Requirement already satisfied: six>=1.5 in c:\users\chint\anaconda3\lib\s
ite-packages (from python-dateutil>=2.8.1->pandas) (1.16.0)
```

In [2]:

```
!pip install --upgrade scikit-learn -q --user
```

In [3]:

```
folder_path = 'F:/MSc Data Science MMU/Semister 3/Project/THESIS/Footb
# Grouped the appropriate csv files
csv_files = ['0506E0.csv', '0506E1.csv', '0506E2.csv', '0506E3.csv', '
             '0607E0.csv', '0607E1.csv', '0607E2.csv', '0607E3.csv', '0
             '0708E0.csv', '0708E1.csv', '0708E2.csv', '0708E3.csv', '0
             '0809E0.csv', '0809E1.csv', '0809E2.csv', '0809E3.csv', '0
             '0910E0.csv', '0910E1.csv', '0910E2.csv', '0910E3.csv', '0
             '1011E0.csv', '1011E1.csv', '1011E2.csv', '1011E3.csv', '1
             '1112E0.csv', '1112E1.csv', '1112E2.csv', '1112E3.csv', '1
             '1213E0.csv', '1213E1.csv', '1213E2.csv', '1213E3.csv', '1
             '1314E0.csv', '1314E1.csv', '1314E2.csv', '1314E3.csv', '1
             '1415E0.csv', '1415E1.csv', '1415E2.csv', '1415E3.csv', '1
             '1516E0.csv', '1516E1.csv', '1516E2.csv', '1516E3.csv', '1
             '1617E0.csv', '1617E1.csv', '1617E2.csv', '1617E3.csv', '1
             '1718E0.csv', '1718E1.csv', '1718E2.csv', '1718E3.csv', '1
             '1819E0.csv', '1819E1.csv', '1819E2.csv', '1819E3.csv', '1
             '1920E0.csv', '1920E1.csv', '1920E2.csv', '1920E3.csv', '1
             '2021E0.csv', '2021E1.csv', '2021E2.csv', '2021E3.csv', '2
             '2122E0.csv', '2122E1.csv', '2122E2.csv', '2122E3.csv', '2
             '2223E0.csv', '2223E1.csv', '2223E2.csv', '2223E3.csv', '2

```

In [4]:
```
file_paths = [folder_path + '/' + file for file in csv_files]
```

In [5]:
```
# Load and concatenate the CSV files into a single DataFrame
fbdata = pd.concat([pd.read_csv(file, encoding='cp1252') for file in f

```

2. DATA EXPLORATION

In [6]:
```
fbdata.head(5)
```

Out[6]:

	Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	...	Avg
0	E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	...	
1	E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	...	
2	E0	13/08/05	Fulham	Birmingham	0.0	0.0	D	0.0	0.0	D	...	
3	E0	13/08/05	Man City	West Brom	0.0	0.0	D	0.0	0.0	D	...	
4	E0	13/08/05	Middlesbrough	Liverpool	0.0	0.0	D	0.0	0.0	D	...	

5 rows × 139 columns

In [7]:
```
num_rows = fbdata.shape[0]
print("Numbber of rows:", num_rows)
```

```
Numbber of rows: 45969
```

In [8]:
```
div_types = fbdata['Div'].unique()
print("Unique types in 'Div' column:", div_types)
```

```
Unique types in 'Div' column: ['E0' 'E1' 'E2' 'E3' 'EC' nan]
```

In [9]:
```
missing_values = fbdata['Div'].isna().sum()
```

In [10]:
```
fbdata.dropna(subset=['Div'], inplace=True)
```

In [11]:
```
fbdata.set_index('Div', inplace=True)
fbdata.head(20)
```

Out[11]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Refe
E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	M R
E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	G I
E0	13/08/05	Fulham	Birmingham	0.0	0.0	D	0.0	0.0	D	R Sty
E0	13/08/05	Man City	West Brom	0.0	0.0	D	0.0	0.0	D	C I
E0	13/08/05	Middlesbrough	Liverpool	0.0	0.0	D	0.0	0.0	D	M Hal
E0	13/08/05	Portsmouth	Tottenham	0.0	2.0	A	0.0	1.0	A	B Kni
E0	13/08/05	Sunderland	Charlton	1.0	3.0	A	1.0	1.0	D	H We
E0	13/08/05	West Ham	Blackburn	3.0	1.0	H	0.0	1.0	A	A W
E0	14/08/05	Arsenal	Newcastle	2.0	0.0	H	0.0	0.0	D	S Benr
E0	14/08/05	Wigan	Chelsea	0.0	1.0	A	0.0	0.0	D	Clattenb
E0	20/08/05	Birmingham	Man City	1.0	2.0	A	1.0	1.0	D	Clattenb
E0	20/08/05	Blackburn	Fulham	2.0	1.0	H	1.0	0.0	H	H We
E0	20/08/05	Charlton	Wigan	1.0	0.0	H	1.0	0.0	H	R Sty
E0	20/08/05	Liverpool	Sunderland	1.0	0.0	H	1.0	0.0	H	B Kni
E0	20/08/05	Man United	Aston Villa	1.0	0.0	H	0.0	0.0	D	P Do
E0	20/08/05	Newcastle	West Ham	0.0	0.0	D	0.0	0.0	D	Gallag
E0	20/08/05	Tottenham	Middlesbrough	2.0	0.0	H	0.0	0.0	D	M Atkins
E0	20/08/05	West Brom	Portsmouth	2.0	1.0	H	1.0	0.0	H	M R
E0	21/08/05	Bolton	Everton	0.0	1.0	A	0.0	0.0	D	A W
E0	21/08/05	Chelsea	Arsenal	1.0	0.0	H	0.0	0.0	D	G I

20 rows × 138 columns

In [12]:
```
#fbdata= fbdata[fbdata.index !='E0']
```

In [13]:
```
1 fbdata.tail(5)
```

Out[13]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee
EC	29/04/2023	Notts County	York	1.0	1.0	D	1.0	0.0	H	G Rollason
EC	29/04/2023	Oldham	Bromley	1.0	1.0	D	1.0	0.0	H	R Watkins
EC	29/04/2023	Solihull	Woking	0.0	1.0	A	0.0	0.0	D	A Bannister
EC	29/04/2023	Southend	Wealdstone	2.0	1.0	H	0.0	1.0	A	G Parsons
EC	29/04/2023	Torquay	Wrexham	1.0	1.0	D	0.0	0.0	D	G Rhodes

5 rows × 138 columns

In [14]:
```
1 fbdata.count()
```

Out[14]:
```
Date        45952
HomeTeam    45952
AwayTeam    45952
FTHG        45952
FTAG        45952
            ...
PCAHA        9841
MaxCAHH      9856
MaxCAHA      9856
AvgCAHH      9856
AvgCAHA      9856
Length: 138, dtype: int64
```

In [15]:
```
1 # Identify the data types
2 fbdata.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 45952 entries, E0 to EC
Columns: 138 entries, Date to AvgCAHA
dtypes: float64(130), object(8)
memory usage: 48.7+ MB
```

It revals 45969 rows of data belongs to E0, E1, E2, E3, EC EPL Football Divisions spreaded through 138 features from Date to AvgCAHA, where 130 features are of floats and 8 objects in the dataset.

In [16]:
```
1 fbdata.head(2)
```

Out[16]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	...
E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	M Riley	...
E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	G Poll	...

2 rows × 138 columns

In [17]:
```
1 # See the list of features in the data set
2 all_features = fbdata.columns.tolist()
3 print(all_features)
```

```
['Date', 'HomeTeam', 'AwayTeam', 'FTHG', 'FTAG', 'FTR', 'HTHG', 'HTAG',
'HTR', 'Referee', 'HS', 'AS', 'HST', 'AST', 'HF', 'AF', 'HC', 'AC', 'HY',
'AY', 'HR', 'AR', 'B365H', 'B365D', 'B365A', 'BWH', 'BWD', 'BWA', 'GBH',
'GBD', 'GBA', 'IWH', 'IWD', 'IWA', 'LBH', 'LBD', 'LBA', 'SBH', 'SBD', 'SB
A', 'WHH', 'WHD', 'WHA', 'SJH', 'SJD', 'SJA', 'VCH', 'VCD', 'VCA', 'Bb1X
2', 'BbMxH', 'BbAvH', 'BbMxD', 'BbAvD', 'BbMxA', 'BbAvA', 'BbOU', 'BbMx>
2.5', 'BbAv>2.5', 'BbMx<2.5', 'BbAv<2.5', 'BbAH', 'BbAHh', 'BbMxAHH', 'Bb
AvAHH', 'BbMxAHA', 'BbAvAHA', 'BSH', 'BSD', 'BSA', 'PSH', 'PSD', 'PSA',
'PSCH', 'PSCD', 'PSCA', 'Time', 'MaxH', 'MaxD', 'MaxA', 'AvgH', 'AvgD',
'AvgA', 'B365>2.5', 'B365<2.5', 'P>2.5', 'P<2.5', 'Max>2.5', 'Max<2.5',
'Avg>2.5', 'Avg<2.5', 'AHh', 'B365AHH', 'B365AHA', 'PAHH', 'PAHA', 'MaxAH
H', 'MaxAHA', 'AvgAHH', 'AvgAHA', 'B365CH', 'B365CD', 'B365CA', 'BWCH',
'BWCD', 'BWCA', 'IWCH', 'IWCD', 'IWCA', 'WHCH', 'WHCD', 'WHCA', 'VCCH',
'VCCD', 'VCCA', 'MaxCH', 'MaxCD', 'MaxCA', 'AvgCH', 'AvgCD', 'AvgCA', 'B3
65C>2.5', 'B365C<2.5', 'PC>2.5', 'PC<2.5', 'MaxC>2.5', 'MaxC<2.5', 'AvgC>
2.5', 'AvgC<2.5', 'AHCh', 'B365CAHH', 'B365CAHA', 'PCAHH', 'PCAHA', 'MaxC
AHH', 'MaxCAHA', 'AvgCAHH', 'AvgCAHA']
```

In [18]:
```
1 # fig, chart = plt.subplots()
2 # data = fbdata['FTR'].value_counts()points =fbdata.index
3 # frequency = fbdata. valueschart.bar(points,
4 # frequency)chart.set.title('Frequency of EPL Leage results205-2022')
5 # chart.set_xlabel('Results Type')
6 # chart.set_ylabel('Frequency')
```

2.1. Summary Statistics

In [82]:
```
#Check summary statistics of belived to be influentoial data
summary_stats = fbdata[['FTHG', 'FTAG', 'HTHG', 'HTAG', 'HS', 'AS', 'H
summary_stats = summary_stats.round(1)
print(summary_stats)
```

```
          FTHG      FTAG      HTHG      HTAG        HS        AS       HST
AST  \
count  45952.0  45952.0  45951.0  45951.0  42309.0  42309.0  42309.0  423
09.0
mean       1.4      1.2      0.6      0.5     12.3     10.1      5.3
4.3
std        1.2      1.1      0.8      0.7      4.8      4.3      2.9
2.5
min        0.0      0.0      0.0      0.0      0.0      0.0      0.0
0.0
25%        1.0      0.0      0.0      0.0      9.0      7.0      3.0
2.0
50%        1.0      1.0      0.0      0.0     12.0     10.0      5.0
4.0
75%        2.0      2.0      1.0      1.0     15.0     13.0      7.0
6.0
max        9.0      9.0      7.0      5.0     43.0     35.0     24.0
20.0

            HF       AF  ...  AvgC<2.5    AHCh  B365CAHH  B365CAHA   PCAH
H  \
count  42307.0  42307.0  ...    9856.0  9856.0    9839.0    9839.0  9841.
0
mean      11.1     11.6  ...       1.9    -0.2       1.9       1.9     1.
9
std        3.7      3.8  ...       0.3     0.6       0.1       0.1     0.
1
min        0.0      0.0  ...       1.4    -3.8       1.6       1.3     1.
6
25%        9.0      9.0  ...       1.7    -0.5       1.8       1.8     1.
9
50%       11.0     11.0  ...       1.8    -0.2       1.9       1.9     1.
9
75%       13.0     14.0  ...       2.0     0.2       2.0       2.0     2.
0
max       33.0     77.0  ...       5.8     3.0       3.6       2.4     3.
9

        PCAHA  MaxCAHH  MaxCAHA  AvgCAHH  AvgCAHA
count  9841.0   9856.0   9856.0   9856.0   9856.0
mean      2.0      2.0      2.0      1.9      1.9
std       0.1      0.1      0.1      0.1      0.1
min       1.3      1.7      1.3      1.7      1.3
25%       1.9      1.9      1.9      1.8      1.8
50%       1.9      2.0      2.0      1.9      1.9
75%       2.0      2.1      2.1      2.0      2.0
max       2.4      4.0      2.5      3.7      2.2

[8 rows x 130 columns]
```

In [83]:
```
#Check summary statistics of belived to be influentoial data
summary_stats = fbdata[['Date', 'FTR', 'FTHG', 'FTAG', 'HTHG', 'HTAG',
summary_stats = summary_stats.round(1)
print(summary_stats)
```

```
            FTHG      FTAG      HTHG      HTAG        HS        AS       HST
AST   \
count   45952.0   45952.0   45951.0   45951.0   42309.0   42309.0   42309.0   423
09.0
mean        1.4       1.2       0.6       0.5      12.3      10.1       5.3
4.3
std         1.2       1.1       0.8       0.7       4.8       4.3       2.9
2.5
min         0.0       0.0       0.0       0.0       0.0       0.0       0.0
0.0
25%         1.0       0.0       0.0       0.0       9.0       7.0       3.0
2.0
50%         1.0       1.0       0.0       0.0      12.0      10.0       5.0
4.0
75%         2.0       2.0       1.0       1.0      15.0      13.0       7.0
6.0
max         9.0       9.0       7.0       5.0      43.0      35.0      24.0
20.0

             HF        AS        HF        AF        HC        AC        HY
AY   \
count   42307.0   42309.0   42307.0   42307.0   42309.0   42309.0   45951.0   459
51.0
mean       11.1      10.1      11.1      11.6       5.8       4.8       1.4
1.7
std         3.7       4.3       3.7       3.8       2.9       2.6       1.2
1.3
min         0.0       0.0       0.0       0.0       0.0       0.0       0.0
0.0
25%         9.0       7.0       9.0       9.0       4.0       3.0       0.0
1.0
50%        11.0      10.0      11.0      11.0       6.0       5.0       1.0
2.0
75%        13.0      13.0      13.0      14.0       8.0       6.0       2.0
2.0
max        33.0      35.0      33.0      77.0      24.0      21.0      11.0
9.0

             HR        AR
count   45951.0   45950.0
mean        0.1       0.1
std         0.3       0.3
min         0.0       0.0
25%         0.0       0.0
50%         0.0       0.0
75%         0.0       0.0
max         3.0       3.0
```

2.2. Performance Analysis of teams

In [ ]:

```
team_performance = fbdata.groupby('HomeTeam').mean()[['FTHG', 'FTAG',
#team_operformance_columns= ['Team', 'AvgGoals', 'AvgShots', 'AvgShots
print(team_performance)
```

In [73]:

```
team_performance = fbdata.groupby('HomeTeam').mean()[['FTHG', 'FTAG',
#team_operformance_columns= ['Team', 'AvgGoals', 'AvgShots', 'AvgShots
print(team_performance)
```

```
                HomeTeam      FTHG      FTAG      HTHG      HTAG         H
S  \
0    AFC Telford United  1.043478  1.130435  0.347826  0.521739   8.56521
7
1         AFC Wimbledon  1.329114  1.265823  0.582278  0.556962  11.25949
4
2            Accrington  1.534483  1.330049  0.662562  0.544335  12.48768
5
3             Aldershot  1.395062  1.232099  0.632099  0.570370   9.07171
3
4         Alfreton Town  1.717391  1.739130  0.869565  0.750000  11.05434
8
..                  ...       ...       ...       ...       ...
...
154             Wolves  1.379581  1.162304  0.638743  0.518325  12.61256
5
155            Wrexham  1.586207  0.903941  0.753695  0.413793   9.41666
7
156            Wycombe  1.344743  1.095355  0.633252  0.444988  10.85085
6
157             Yeovil  1.144963  1.184275  0.538084  0.486486   9.40372
7
158               York  1.334459  1.097973  0.611486  0.449324  11.97600
0

           AS       HST       AST         HF         AS         HF  \
0    10.347826  4.913043  5.565217  11.913043  10.347826  11.913043
1     9.962025  4.481013  4.075949  11.734177   9.962025  11.734177
2    10.167488  5.371921  4.386700  10.906404  10.167488  10.906404
3     6.585657  4.290837  3.159363  10.462151   6.585657  10.462151
4    11.217391  5.489130  6.032609  11.097826  11.217391  11.097826
..         ...       ...       ...        ...        ...        ...
154  10.900524  5.277487  4.730366  11.295812  10.900524  11.295812
155   6.265873  4.178571  2.821429  11.190476   6.265873  11.190476
156   8.540342  4.400978  3.481663  12.078240   8.540342  12.078240
157   8.863354  4.133540  3.835404  11.593168   8.863354  11.593168
158   8.852000  5.820000  4.220000  10.076000   8.852000  10.076000

           AF        HC        AC        HY        AY        HR         A
R
0    11.217391  4.956522  4.869565  2.086957  1.826087  0.086957  0.21739
1
1    11.000000  5.620253  4.803797  1.354430  1.734177  0.041139  0.12341
8
2    12.421182  5.512315  5.128079  1.413793  1.467980  0.110837  0.09852
2
3    10.697211  5.601594  4.784861  1.392593  1.735802  0.074074  0.10123
5
4    10.815217  5.673913  5.478261  1.402174  1.500000  0.130435  0.10869
6
..         ...       ...       ...       ...       ...       ...
...
154  10.900524  5.952880  4.643979  1.353403  1.845550  0.047120  0.09162
3
155  11.940476  5.789683  4.686508  1.401478  1.709360  0.088670  0.10098
5
156  12.029340  5.836186  4.728606  1.359413  1.809291  0.063570  0.10513
4
157  11.527950  5.257764  5.270186  1.584767  1.776413  0.103194  0.12285
0
158  11.668000  6.036000  4.496000  1.236486  1.706081  0.060811  0.07770
```

```
3

[159 rows x 19 columns]
```

In [22]:

```
team_performance = fbdata.groupby('AwayTeam').mean()[['FTAG', 'AS', 'A
#team_operformance_columns= ['Team', 'AvgGoals', 'AvgShots', 'AvgShots
print(team_performance)
```

```
              AwayTeam      FTAG         AS       AST
0    AFC Telford United  0.913043   8.043478  3.347826
1         AFC Wimbledon  1.073016   9.625397  3.926984
2            Accrington  1.073710  10.496314  4.255528
3             Aldershot  1.176904   9.466135  4.677291
4         Alfreton Town  0.989130   7.869565  3.760870
..                  ...       ...        ...       ...
154             Wolves  1.152231  10.425197  4.301837
155            Wrexham  1.216749   9.472222  4.722222
156            Wycombe  1.208845  10.611794  4.343980
157             Yeovil  1.068796   9.338509  4.288820
158               York  1.209459   8.908000  4.192000

[159 rows x 4 columns]
```

2.3. Home vs Away Performance

In [23]:

```
home_away_performance = fbdata.groupby('HomeTeam')[['FTHG', 'HS', 'HST
away_performance = fbdata.groupby('AwayTeam')[['FTAG', 'AS', 'AST']].m
home_away_performance.columns = ['Team', 'AvgHomeGoals', 'AvgHomeShots
away_performance.columns = ['Team', 'AvgAwayGoals', 'AvgAwayShots', 'A
performance_comparison = home_away_performance.merge(away_performance,
print(performance_comparison)

```

```
                  Team  AvgHomeGoals  AvgHomeShots  AvgHomeShotsOnTarget
\
0    AFC Telford United      1.043478      8.565217              4.913043
1         AFC Wimbledon      1.329114     11.259494              4.481013
2            Accrington      1.534483     12.487685              5.371921
3             Aldershot      1.395062      9.071713              4.290837
4         Alfreton Town      1.717391     11.054348              5.489130
..                  ...           ...           ...                   ...
154             Wolves      1.379581     12.612565              5.277487
155            Wrexham      1.586207      9.416667              4.178571
156            Wycombe      1.344743     10.850856              4.400978
157             Yeovil      1.144963      9.403727              4.133540
158               York      1.334459     11.976000              5.820000

     AvgAwayGoals  AvgAwayShots  AvgAwayShotsOnTarget
0        0.913043      8.043478              3.347826
1        1.073016      9.625397              3.926984
2        1.073710     10.496314              4.255528
3        1.176904      9.466135              4.677291
4        0.989130      7.869565              3.760870
..            ...           ...                   ...
154      1.152231     10.425197              4.301837
155      1.216749      9.472222              4.722222
156      1.208845     10.611794              4.343980
157      1.068796      9.338509              4.288820
158      1.209459      8.908000              4.192000

[159 rows x 7 columns]
```

In [86]:

```
home_performance = fbdata.groupby('HomeTeam')[['FTHG', 'HS', 'HST']].m
away_performance = fbdata.groupby('AwayTeam')[['FTAG', 'AS', 'AST']].m
home_away_performance.columns = ['Team', 'AvgHomeGoals', 'AvgHomeShots
away_performance.columns = ['Team', 'AvgHomeGoals', 'AvgHomeShots', 'A
performance_comparision = home_away_performance.merge(away_performance
print(performance_comparision)
```

```
                  Team  AvgHomeGoals_x  AvgHomeShots_x  \
0    AFC Telford United        1.043478        8.565217
1         AFC Wimbledon        1.329114       11.259494
2            Accrington        1.534483       12.487685
3             Aldershot        1.395062        9.071713
4         Alfreton Town        1.717391       11.054348
..                  ...             ...             ...
154             Wolves        1.379581       12.612565
155            Wrexham        1.586207        9.416667
156            Wycombe        1.344743       10.850856
157             Yeovil        1.144963        9.403727
158               York        1.334459       11.976000

     AvgHomeShotsOnTarget_x  AvgHomeGoals_y  AvgHomeShots_y  \
0                  4.913043        0.913043        8.043478
1                  4.481013        1.073016        9.625397
2                  5.371921        1.073710       10.496314
3                  4.290837        1.176904        9.466135
4                  5.489130        0.989130        7.869565
..                      ...             ...             ...
154                5.277487        1.152231       10.425197
155                4.178571        1.216749        9.472222
156                4.400978        1.208845       10.611794
157                4.133540        1.068796        9.338509
158                5.820000        1.209459        8.908000

     AvgHomeShotsOnTarget_y
0                  3.347826
1                  3.926984
2                  4.255528
3                  4.677291
4                  3.760870
..                      ...
154                4.301837
155                4.722222
156                4.343980
157                4.288820
158                4.192000

[159 rows x 7 columns]
```

In [25]:
```
1 fbdata.head(2)
```

Out[25]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	...
E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	M Riley	...
E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	G Poll	...

2 rows × 138 columns

2.4. Win loss Analysis

In [26]:
```
1 outcome_frequency = fbdata['FTR'].value_counts()
2 print(outcome_frequency)
```

```
H    19936
A    13892
D    12124
Name: FTR, dtype: int64
```

2.5. Corelation analysis

In [27]:
```
correlation_matrix = fbdata[['FTHG', 'FTAG', 'HS', 'AS', 'HST', 'AST',
print(correlation_matrix)
```

```
          FTHG      FTAG        HS        AS       HST       AST        H
F  \
FTHG  1.000000 -0.033819  0.255356 -0.063088  0.433339 -0.050687 -0.03298
9
FTAG -0.033819  1.000000 -0.056333  0.258942 -0.055908  0.436239 -0.01597
5
HS    0.255356 -0.056333  1.000000 -0.107814  0.655372 -0.086628 -0.08417
2
AS   -0.063088  0.258942 -0.107814  1.000000 -0.097892  0.660412  0.03885
1
HST   0.433339 -0.055908  0.655372 -0.097892  1.000000  0.062756 -0.04568
0
AST  -0.050687  0.436239 -0.086628  0.660412  0.062756  1.000000  0.02719
4
HF   -0.032989 -0.015975 -0.084172  0.038851 -0.045680  0.027194  1.00000
0
AF   -0.041507 -0.013717  0.010856 -0.059030  0.010993 -0.032117  0.17004
2
HC    0.000491 -0.013617  0.381839 -0.206187  0.279116 -0.096472 -0.09429
0
AC   -0.026352  0.010776 -0.206259  0.366994 -0.108994  0.265034 -0.00421
4
HY   -0.061414  0.067959 -0.069493  0.086044 -0.091741  0.042907  0.32363
3
AY    0.013048 -0.007903  0.041946 -0.057443  0.010613 -0.047338  0.07927
0
HR   -0.066732  0.111610 -0.102485  0.098343 -0.070027  0.098565  0.05019
0
AR    0.102782 -0.069834  0.092428 -0.115762  0.092210 -0.082167  0.02944
1

             AF        HC        AC        HY        AY        HR        A
R
FTHG -0.041507  0.000491 -0.026352 -0.061414  0.013048 -0.066732  0.10278
2
FTAG -0.013717 -0.013617  0.010776  0.067959 -0.007903  0.111610 -0.06983
4
HS    0.010856  0.381839 -0.206259 -0.069493  0.041946 -0.102485  0.09242
8
AS   -0.059030 -0.206187  0.366994  0.086044 -0.057443  0.098343 -0.11576
2
HST   0.010993  0.279116 -0.108994 -0.091741  0.010613 -0.070027  0.09221
0
AST  -0.032117 -0.096472  0.265034  0.042907 -0.047338  0.098565 -0.08216
7
HF    0.170042 -0.094290 -0.004214  0.323633  0.079270  0.050190  0.02944
1
AF    1.000000 -0.027666 -0.065905  0.070192  0.342220  0.031469  0.06641
0
HC   -0.027666  1.000000 -0.187415 -0.065869  0.028184 -0.062552  0.05700
0
AC   -0.065905 -0.187415  1.000000  0.030230 -0.057771  0.052568 -0.07113
1
HY    0.070192 -0.065869  0.030230  1.000000  0.180687  0.078619  0.05070
1
AY    0.342220  0.028184 -0.057771  0.180687  1.000000  0.051188  0.07601
7
HR    0.031469 -0.062552  0.052568  0.078619  0.051188  1.000000  0.05676
9
```

```
AR    0.066410  0.057000 -0.071131  0.050701  0.076017  0.056769  1.00000
0
```

2.6. Time series analysis

In [28]:
```
# fbdata['Date'] = pd.to_datetime(fbdata['Date'])
# fbdata.set_index('Date', inplace=True)
# goals_over_time = fbdata.resample('Y').sum()[['FTHG', 'FTAG']]
# print(goals_over_time)
```

In [29]:
```
# Reset the index
#fbdata.reset_index(inplace=True)
```

2.7. Referee. Analysis

In [30]:
```
Referee_statistics = fbdata.groupby('Referee').mean()[['HF', 'AF', 'HY
print(Referee_statistics)
```

```
                    HF         AF        HY        AY      FTHG      FTAG
Referee
A Ajibola          NaN        NaN  1.363636  1.954545  1.545455  1.090909
A Allison    10.500000  11.500000  1.000000  2.000000  0.500000  0.000000
A Backhouse  10.932961  11.145251  1.427136  1.567839  1.482412  1.105528
A Bannister        NaN        NaN  1.400000  1.600000  1.333333  1.266667
A Bates      10.381818  11.650000  0.981818  1.609091  1.386364  0.990909
...                ...        ...       ...       ...       ...       ...
W Barrett    11.875000  13.625000  1.250000  1.250000  1.875000  0.750000
W Cartmel          NaN        NaN  1.000000  1.312500  1.687500  0.937500
W Finnie     10.902913  11.485437  1.392000  1.696000  1.416000  1.208000
k Stroud     12.000000  19.000000  0.000000  2.000000  0.000000  0.000000
l Mason      11.000000  10.000000  4.000000  1.000000  1.000000  1.000000

[480 rows x 6 columns]
```

2.8. Odd Analysis

In [31]:
```
odds_columns = ['B365H', 'BWH', 'GBH']
odds_comparision = fbdata[['FTHG', 'FTAG'] + odds_columns]
print(odds_comparision)
```

```
      FTHG  FTAG  B365H   BWH   GBH
Div
E0     2.0   2.0   2.30  2.10  2.25
E0     0.0   2.0   5.00  4.35  4.25
E0     0.0   0.0   2.37  2.30  2.30
E0     0.0   0.0   1.72  1.65  1.73
E0     0.0   0.0   2.87  2.90  2.70
..     ...   ...    ...   ...   ...
EC     1.0   1.0   1.25  1.21   NaN
EC     1.0   1.0   2.70  2.55   NaN
EC     0.0   1.0   3.80  3.50   NaN
EC     2.0   1.0   1.50  1.44   NaN
EC     1.0   1.0   7.50  5.50   NaN

[45952 rows x 5 columns]
```

In [32]:
```
fbdata.head(2)
```

Out[32]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	...
E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	M Riley	...
E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	G Poll	...

2 rows × 138 columns

2.9. Feature importance

In [33]:
```
fbdata.head(2)
```

Out[33]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	...
E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	M Riley	...
E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	G Poll	...

2 rows × 138 columns

In [34]:
```
importance_scores = fbdata[['FTHG','FTAG', 'HS', 'AS', 'HST', 'AST', '
print(importance_scores)
```

```
FTHG     1.000000
HST      0.433339
HS       0.255356
B365H    0.197752
BWH      0.196236
GBH      0.187224
AS       0.063088
AST      0.050687
FTAG     0.033819
Name: FTHG, dtype: float64
```

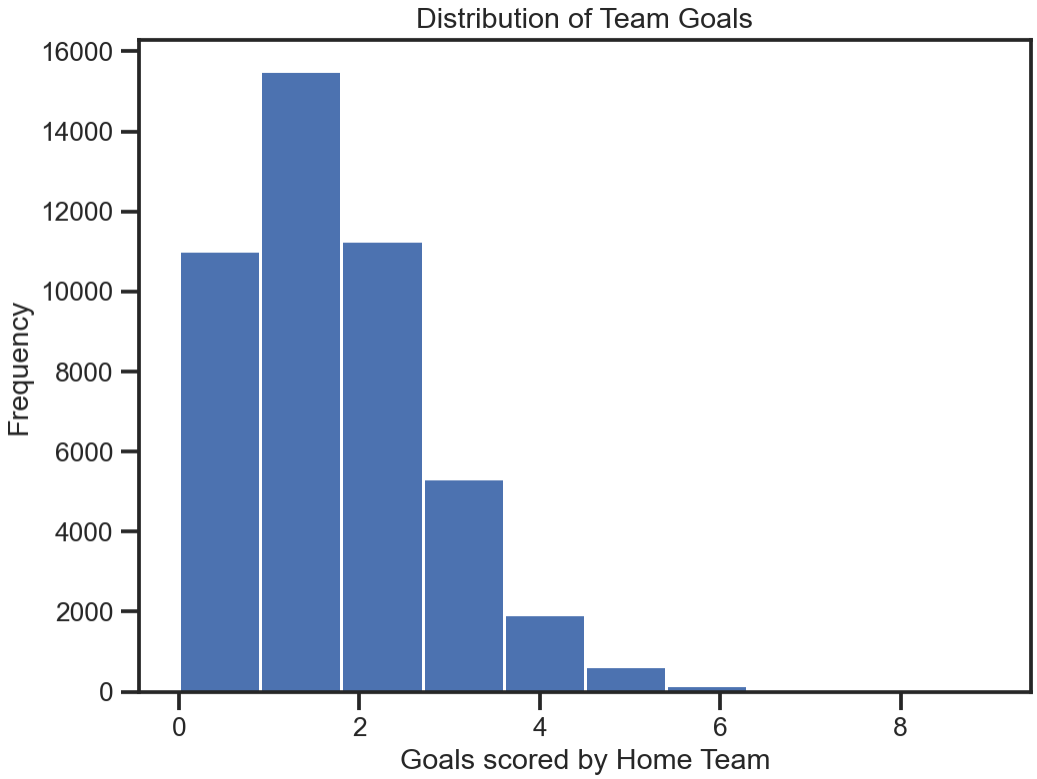
In [35]:
```
importance_scores = fbdata[['FTHG','FTAG', 'HS', 'AS', 'HST', 'AST', '
print(importance_scores)
```

```
FTAG     1.000000
AST      0.436239
AS       0.258942
B365H    0.213356
BWH      0.210991
GBH      0.186701
HS       0.056333
HST      0.055908
FTHG     0.033819
Name: FTAG, dtype: float64
```

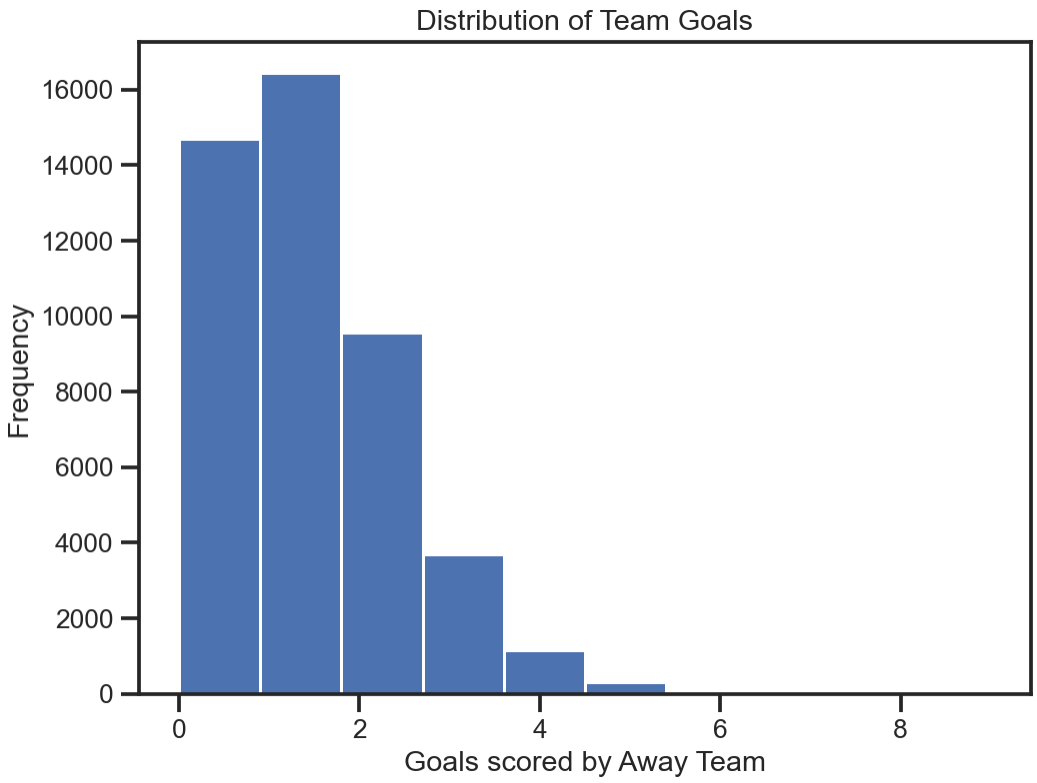
2.10. Visualization

General

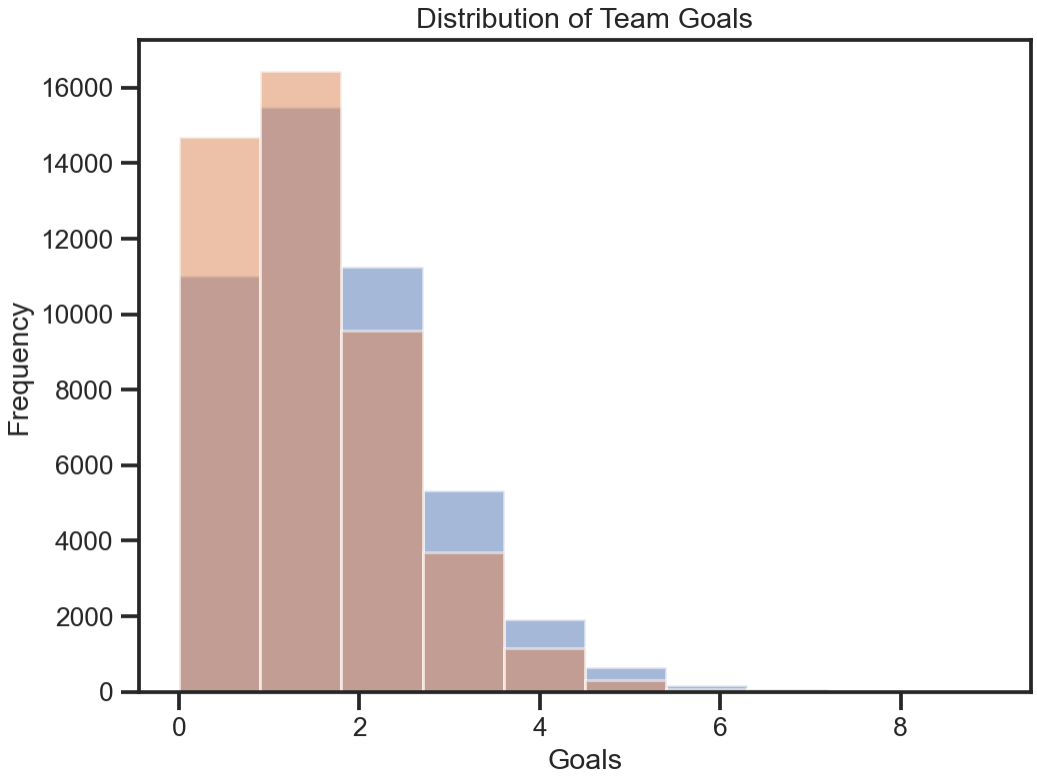
In [36]:
```
fbdata['FTHG'].plot(kind='hist')
plt.xlabel('Goals scored by Home Team')
plt.ylabel('Frequency')
plt.title('Distribution of Team Goals')
plt.show()
```



In [37]:
```
fbdata['FTAG'].plot(kind='hist')
plt.xlabel('Goals scored by Away Team')
plt.ylabel('Frequency')
plt.title('Distribution of Team Goals')
plt.show()
```



In [38]:
```
# plot histogrames for comparision of both
plt.hist(fbdata['FTHG'], bins=10, alpha=0.5, label='Home Team Goal')
plt.hist(fbdata['FTAG'], bins=10, alpha=0.5, label='Away Team Goal')
plt.xlabel('Goals')
plt.ylabel('Frequency')
plt.title('Distribution of Team Goals')
plt.show()
```

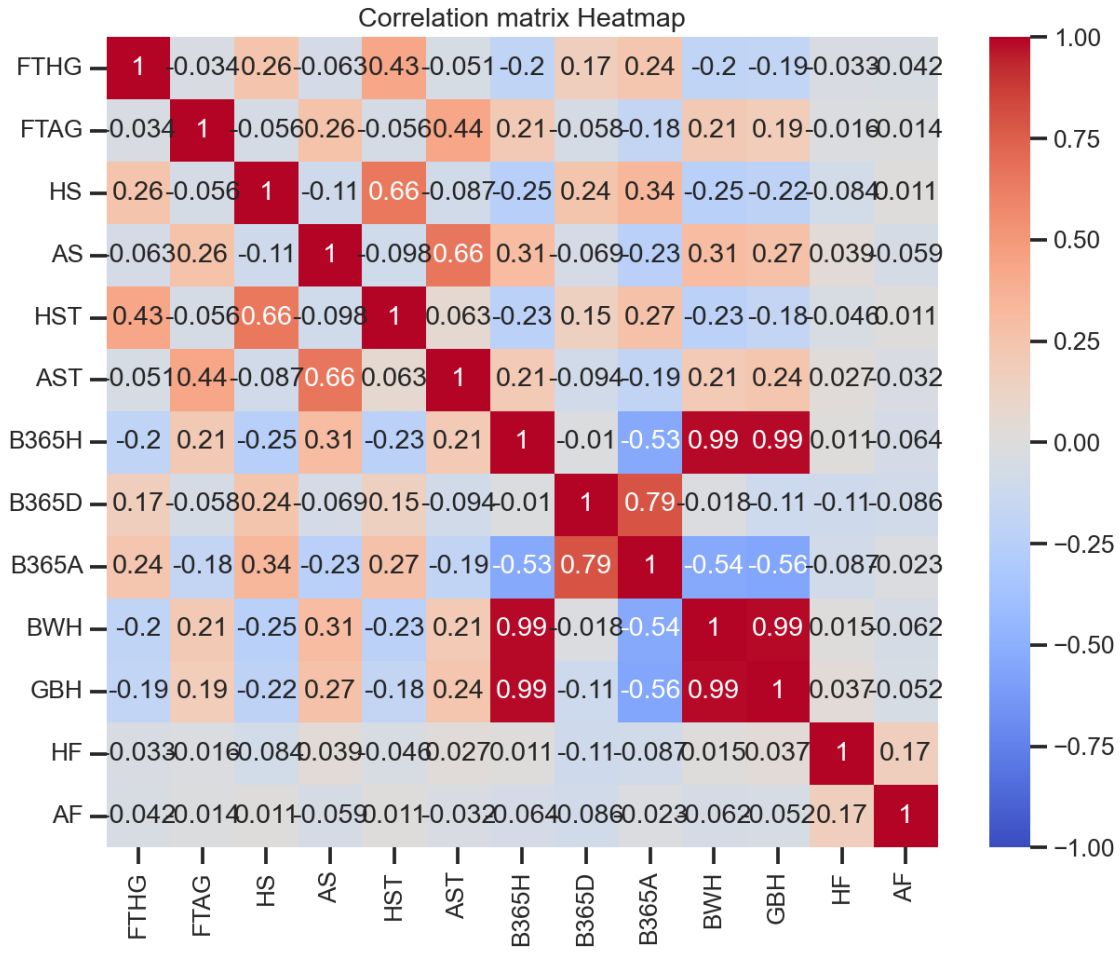


Correlation Matrix

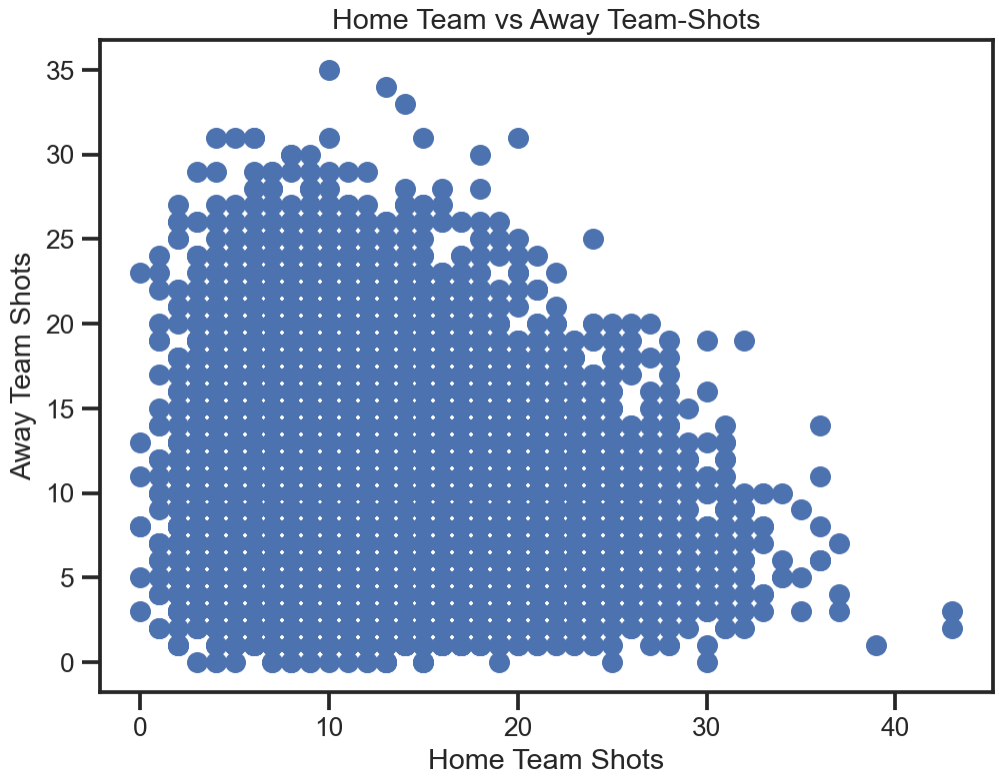
In [39]:

```
# Calculate the correlation matrix
corr_matrix = fbdata[['FTHG','FTAG', 'HS', 'AS', 'HST', 'AST', 'B365H'

# Generate heatmap(anoot=True add the corelation, cmap defines the col
plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', vmin = -1, vmax=
plt.title('Correlation matrix Heatmap')
plt.show()
```



In [40]:
```
# Lets have a scatter plot
x= fbdata['HS']
y= fbdata['AS']
plt.scatter(x,y)
plt.xlabel('Home Team Shots')
plt.ylabel('Away Team Shots')
plt.title('Home Team vs Away Team-Shots')
plt.show()
```



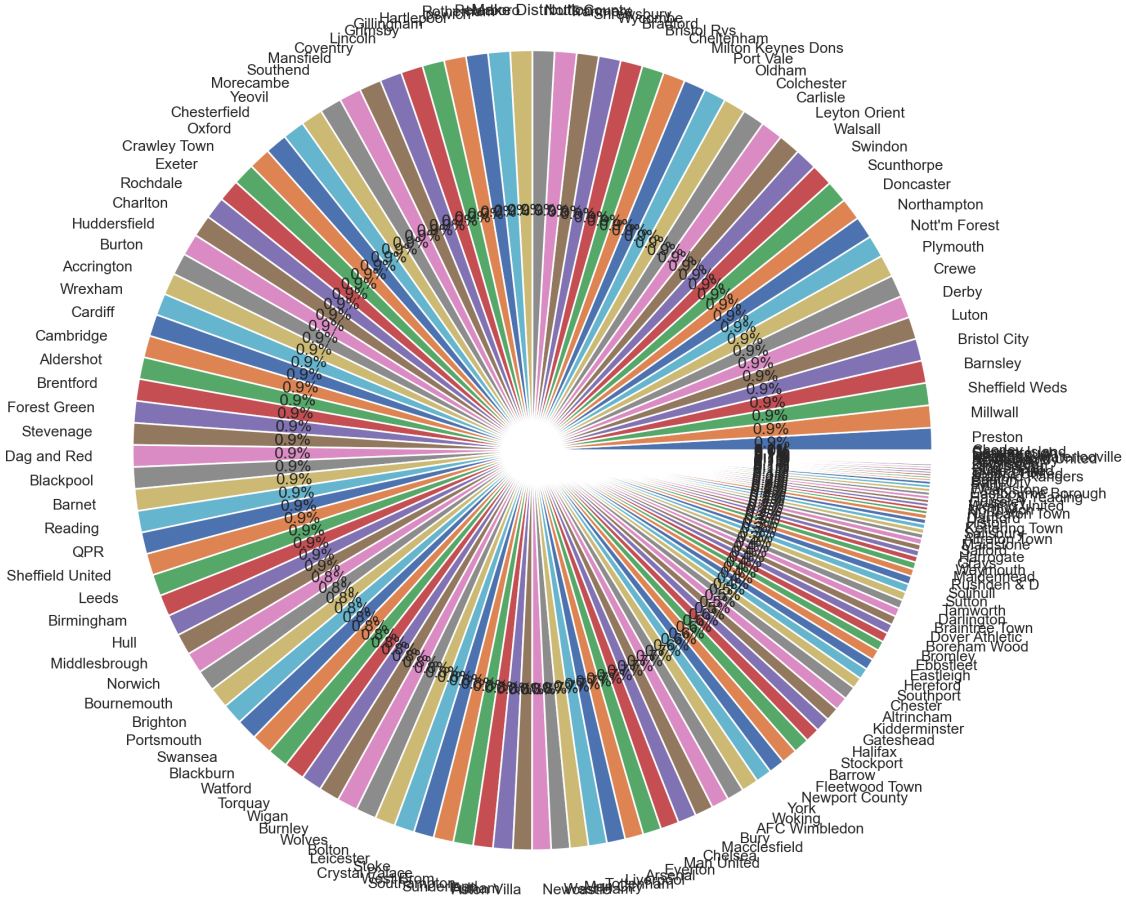
In [ ]:
```
Feature Distribution
```

In [45]:
```
# See the list of features in the data set
all_features = fbdata.columns.tolist()
print(all_features)
```

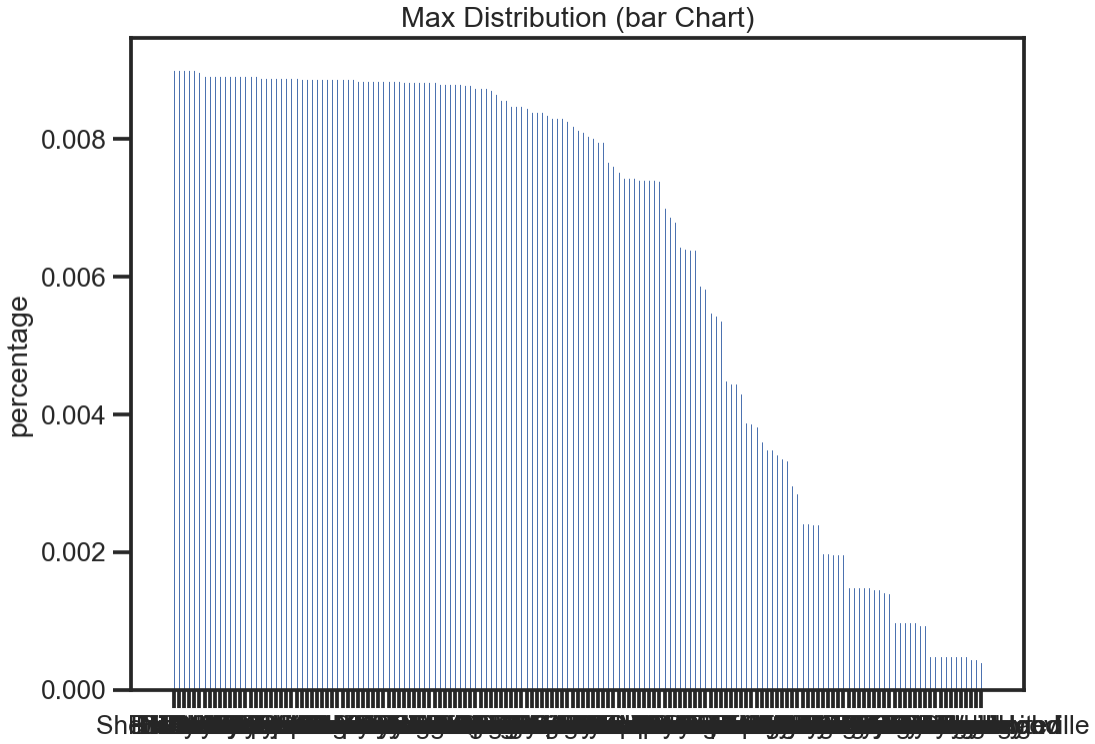
```
['Date', 'HomeTeam', 'AwayTeam', 'FTHG', 'FTAG', 'FTR', 'HTHG', 'HTAG',
'HTR', 'Referee', 'HS', 'AS', 'HST', 'AST', 'HF', 'AF', 'HC', 'AC', 'HY',
'AY', 'HR', 'AR', 'B365H', 'B365D', 'B365A', 'BWH', 'BWD', 'BWA', 'GBH',
'GBD', 'GBA', 'IWH', 'IWD', 'IWA', 'LBH', 'LBD', 'LBA', 'SBH', 'SBD', 'SB
A', 'WHH', 'WHD', 'WHA', 'SJH', 'SJD', 'SJA', 'VCH', 'VCD', 'VCA', 'Bb1X
2', 'BbMxH', 'BbAvH', 'BbMxD', 'BbAvD', 'BbMxA', 'BbAvA', 'BbOU', 'BbMx>
2.5', 'BbAv>2.5', 'BbMx<2.5', 'BbAv<2.5', 'BbAH', 'BbAHh', 'BbMxAHH', 'Bb
AvAHH', 'BbMxAHA', 'BbAvAHA', 'BSH', 'BSD', 'BSA', 'PSH', 'PSD', 'PSA',
'PSCH', 'PSCD', 'PSCA', 'Time', 'MaxH', 'MaxD', 'MaxA', 'AvgH', 'AvgD',
'AvgA', 'B365>2.5', 'B365<2.5', 'P>2.5', 'P<2.5', 'Max>2.5', 'Max<2.5',
'Avg>2.5', 'Avg<2.5', 'AHh', 'B365AHH', 'B365AHA', 'PAHH', 'PAHA', 'MaxAH
H', 'MaxAHA', 'AvgAHH', 'AvgAHA', 'B365CH', 'B365CD', 'B365CA', 'BWCH',
'BWCD', 'BWCA', 'IWCH', 'IWCD', 'IWCA', 'WHCH', 'WHCD', 'WHCA', 'VCCH',
'VCCD', 'VCCA', 'MaxCH', 'MaxCD', 'MaxCA', 'AvgCH', 'AvgCD', 'AvgCA', 'B3
65C>2.5', 'B365C<2.5', 'PC>2.5', 'PC<2.5', 'MaxC>2.5', 'MaxC<2.5', 'AvgC>
2.5', 'AvgC<2.5', 'AHCh', 'B365CAHH', 'B365CAHA', 'PCAHH', 'PCAHA', 'MaxC
AHH', 'MaxCAHA', 'AvgCAHH', 'AvgCAHA']
```

In [62]:

```
fig, ax =plt.subplots()
ax.pie(fbdata['HomeTeam'].value_counts(normalize=True), labels=fbdata[
ax.axis('equal')
ax.set_title('Make Distrbution')
fig.set_size_inches(14,14)
plt.show()
```


Preston
Millwall
Sheffield Weds
Barnsley
Bristol City
Luton
Derby
Crewe
Plymouth
Nott'm Forest
Northampton
Doncaster
Scunthorpe
Swindon
Walsall
Leyton Orient
Carlisle
Colchester
Oldham
Port Vale
Milton Keynes Dons
Cheltenham
Bristol Rvs
Bradford
Wycombe
Shrewsbury
Notts County
Rotherham
Peterboro
Hartlepool
Gillingham
Grimsby
Lincoln
Coventry
Mansfield
Southend
Morecambe
Yeovil
Chesterfield
Oxford
Crawley Town
Exeter
Rochdale
Charlton
Huddersfield
Burton
Accrington
Wrexham
Cardiff
Cambridge
Aldershot
Brentford
Forest Green
Stevenage
Dag and Red
Blackpool
Barnet
Reading
QPR
Sheffield United
Leeds
Birmingham
Hull
Middlesbrough
Norwich
Bournemouth
Brighton
Portsmouth
Swansea
Blackburn
Watford
Torquay
Wigan
Burnley
Wolves
Bolton
Leicester
Stoke
Crystal Palace
Aston Villa
Newcastle
Tottenham
Liverpool
Arsenal
Everton
Man United
Chelsea
Macclesfield
Bury
AFC Wimbledon
Woking
York
Newport County
Fleetwood Town
Barrow
Stockport
Halifax
Gateshead
Kidderminster
Altrincham
Chester
Southport
Hereford
Eastleigh
Ebbsfleet
Bromley
Boreham Wood
Dover Athletic
Braintree Town
Darlington
Tamworth
Sutton
Solihull
Burscough
Whitehead
Weymouth
Harrogate
Gateshead
Stalybridge
Salford
Kettering Town
Histon
Gainsborough
Farsley
Eastbourne Borough
Basingstoke
Aldershot Town
Abingdon United
Hayes & Yeading
Ashford Town
Alfreton
Ilkeston
Bishop's Stortford
Bedfont
Bath City
Bashley
Arlesey Town
Ardley United
Andover
Alvechurch
Ajax
Ajax
Ajax

In [71]:
```
# Convert this to a bar chart
fig2, ax2 = plt.subplots()
ax2.bar(fbdata['HomeTeam'].value_counts().index, fbdata['HomeTeam'].va
ax2.set_ylabel('percentage')
ax2.set_title('Max Distribution (bar Chart)')
fig.set_size_inches(40,40)
plt.show()
```



In [47]:
```
# # Lets visualize numerical features through a histogram
# selected_features =['FTHG', 'FTAG', 'HTHG', 'HTAG', 'HS', 'AS', 'HST

# fig, ax = plt.subplots()
# n, bins, patches = ax.hist(fbdata[selected_features], bins=10, edgec
# ax.set_xlabel('Feature')
# ax.set_ylabel('Frequency')
# ax.set_title('Numerical Feature Distribution')

# # label features
# for i , patch in enumerate(patches):
#     for bar in bars:
#         feature_label= selected_features[i]
#         height = bar.get_height()
#         width = patches.get_widht()
#         x = bar.get_x() + width / 2
#         y= height + 5
#         ax.annotate(feature_label, xy=(x,y), xytext=(0, 5),
#                     textcoords='offset points', ha='center', va='bot

# plt.show()
```

Value counts of features

In [49]:
```
1 fbdata['FTHG'].value_counts()
```

Out[49]:
```
1.0    15501
2.0    11262
0.0    11025
3.0     5333
4.0     1924
5.0      651
6.0      173
7.0       63
8.0       16
9.0        4
Name: FTHG, dtype: int64
```

In [50]:
```
1 fbdata['FTAG'].value_counts()
```

Out[50]:
```
1.0    16436
0.0    14699
2.0     9568
3.0     3697
4.0     1160
5.0      296
6.0       76
7.0       17
9.0        2
8.0        1
Name: FTAG, dtype: int64
```

In [51]:
```
1 fbdata['HST'].value_counts()
```

Out[51]:
```
4.0     6253
5.0     5976
3.0     5627
6.0     5234
2.0     4059
7.0     4028
8.0     2868
1.0     2105
9.0     1966
10.0    1332
11.0     909
0.0      612
12.0     567
13.0     329
14.0     175
15.0     106
16.0      67
17.0      39
18.0      31
19.0      15
20.0       6
21.0       2
24.0       2
22.0       1
Name: HST, dtype: int64
```

In [52]:
```
1 fbdata['B365H'].value_counts()
```

Out[52]:
```
2.10    2280
2.50    2099
2.00    2079
2.20    1805
2.25    1661
        ...
4.25       1
5.80       1
4.90       1
1.38       1
1.05       1
Name: B365H, Length: 142, dtype: int64
```

In [53]:
```
1 fbdata['BWH'].value_counts()
```

Out[53]:
```
2.10    1972
2.00    1909
2.20    1766
2.25    1663
2.30    1574
        ...
6.10       1
7.20       1
1.76       1
1.94       1
6.70       1
Name: BWH, Length: 217, dtype: int64
```

In [54]:
```
1 fbdata.isnull()
```

Out[54]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	...	A
E0	False	False	False	False	False	False	False	False	False	False	...	
E0	False	False	False	False	False	False	False	False	False	False	...	
E0	False	False	False	False	False	False	False	False	False	False	...	
E0	False	False	False	False	False	False	False	False	False	False	...	
E0	False	False	False	False	False	False	False	False	False	False	...	
...	...	...	...	...	...	...	...	...	...	...	...	
EC	False	False	False	False	False	False	False	False	False	False	...	
EC	False	False	False	False	False	False	False	False	False	False	...	
EC	False	False	False	False	False	False	False	False	False	False	...	
EC	False	False	False	False	False	False	False	False	False	False	...	
EC	False	False	False	False	False	False	False	False	False	False	...	

45952 rows × 138 columns

In [55]:
```
1 fbdata.isna().sum()
```

Out[55]:
```
Date            0
HomeTeam        0
AwayTeam        0
FTHG            0
FTAG            0
              ...
PCAHA       36111
MaxCAHH     36096
MaxCAHA     36096
AvgCAHH     36096
AvgCAHA     36096
Length: 138, dtype: int64
```

In [56]:
```
1 # corr_matrix = fbdata.corr()
2 # plt.figure(figsize= (10,8))
3 # sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', linewidths=0.5
4 # plt.title('Feature Corelation Mettrix')
5 # plt.show()
```

In [57]:
```
1 # correlation_threshold = 0.5
2 # potential_components = [component for component in corr_matrix.colum
3 # potential_components.append('Date')
4 # fbdata = fbdata[potential_components]
5 # fbdata
```

In [58]:
```
1 fbdata.head(2)
```

Out[58]:

Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	...
E0	13/08/05	Aston Villa	Bolton	2.0	2.0	D	2.0	2.0	D	M Riley	...
E0	13/08/05	Everton	Man United	0.0	2.0	A	0.0	1.0	A	G Poll	...

2 rows × 138 columns